Concurrency Overview

Example

Transaction 1

```
/* Transfer $100 from Mary to John */
BEGIN TRANSACTION;
    UPDATE Account
    SET balance = balance – 100
    WHERE id = 'Mary';

    UPDATE Account
    SET balance = balance + 100
    WHERE id = 'John';
COMMIT TRANSACTION;
```

Transaction 2

```
/* Get paid interest */
UPDATE Account
SET balance = balance * 1.02;
```

Mary -= 100 John += 100

Mary *= 1.02 John *= 1.02

How are transactions scheduled?

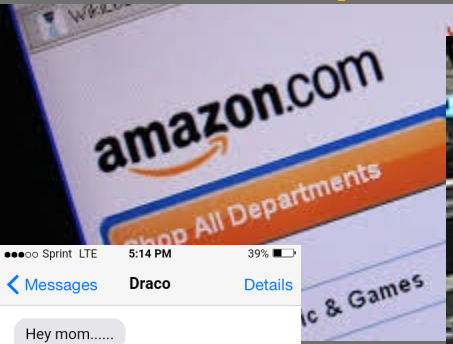


Model assumptions

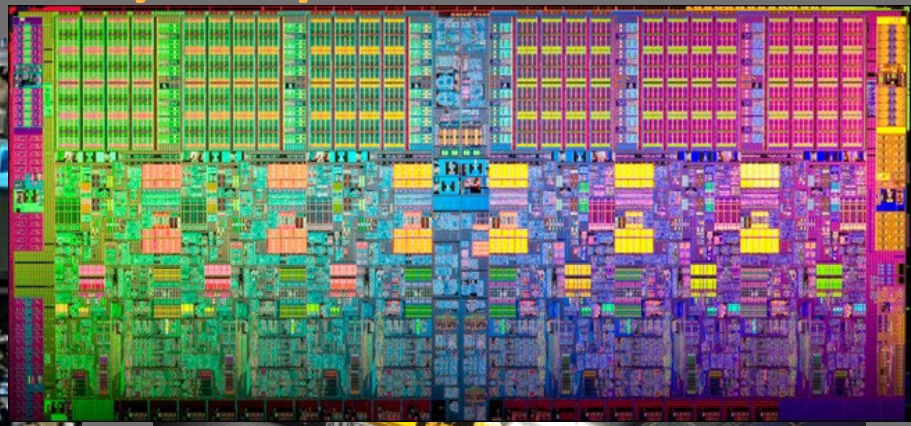
Concurrency

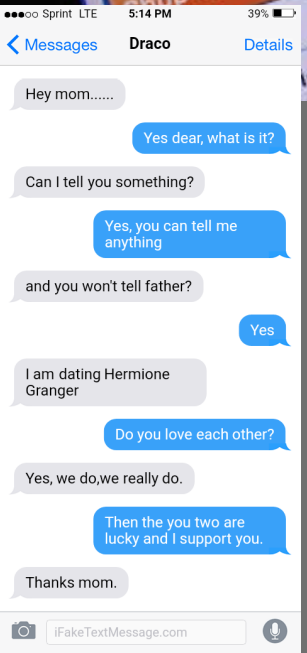
Shared state

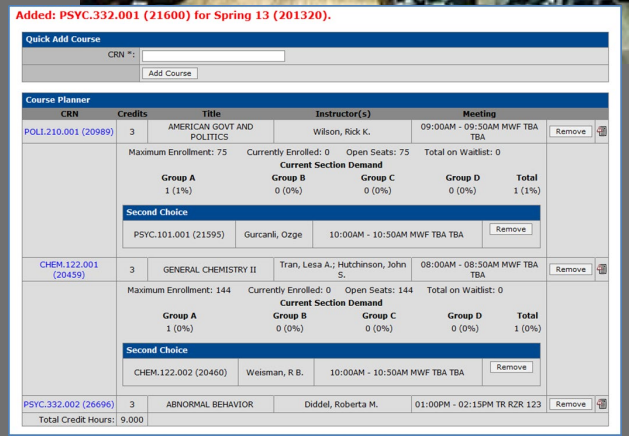
Ubiquitous concurrency & parallelism





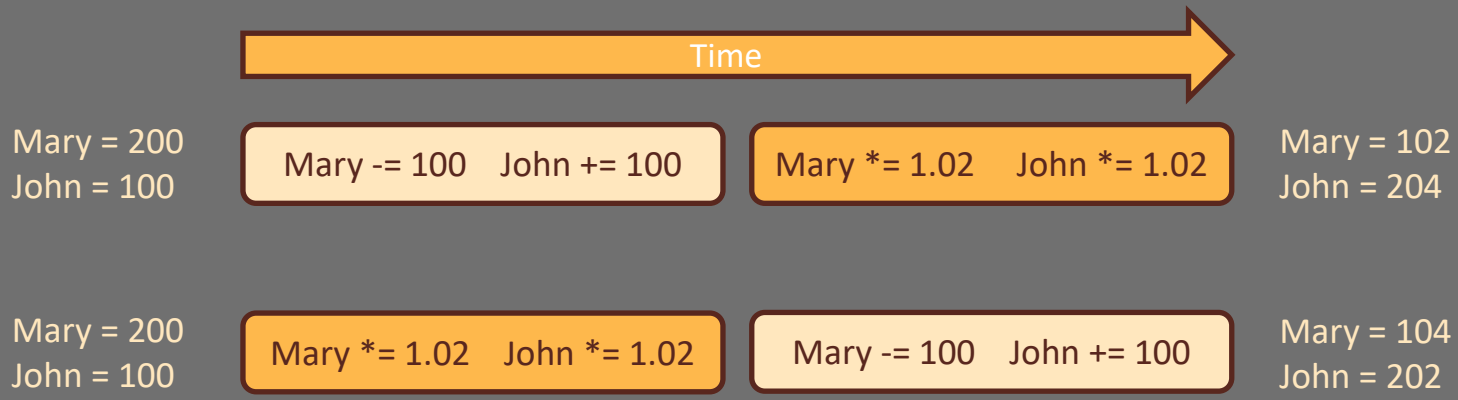






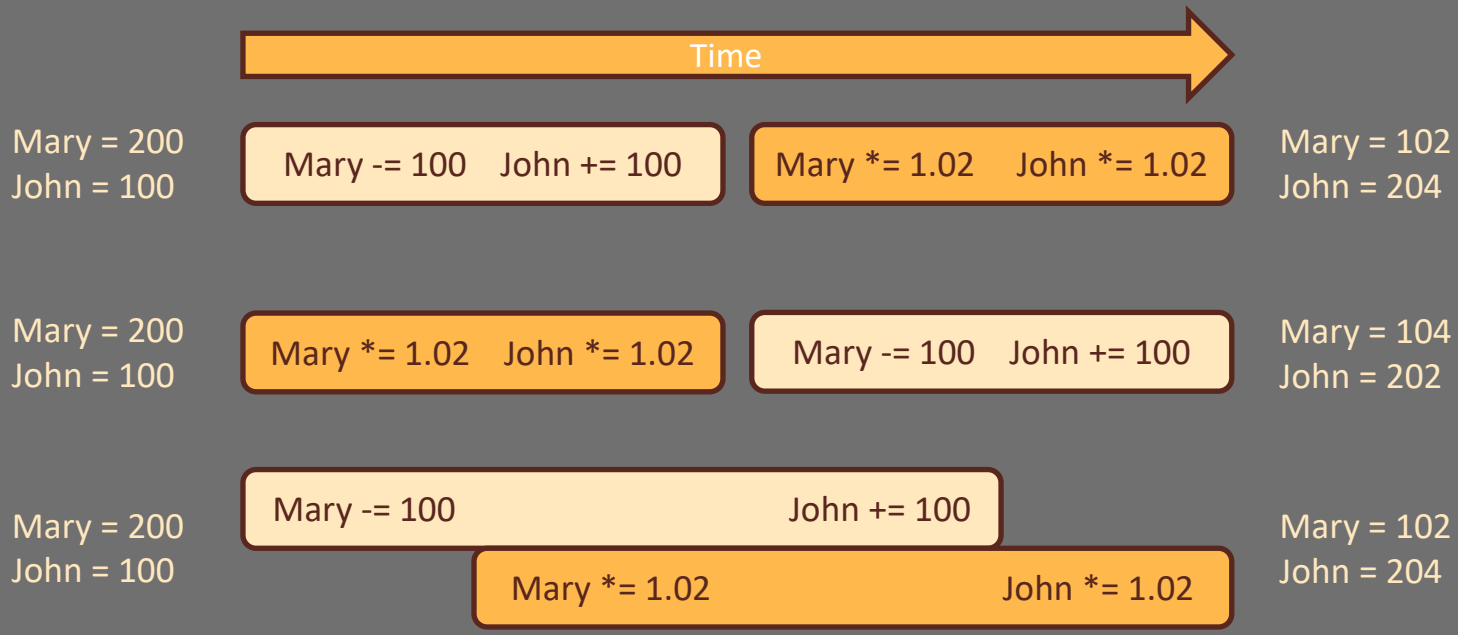


Serial schedules



Isolation: Both serial orderings are considered acceptable.

Serializable schedules



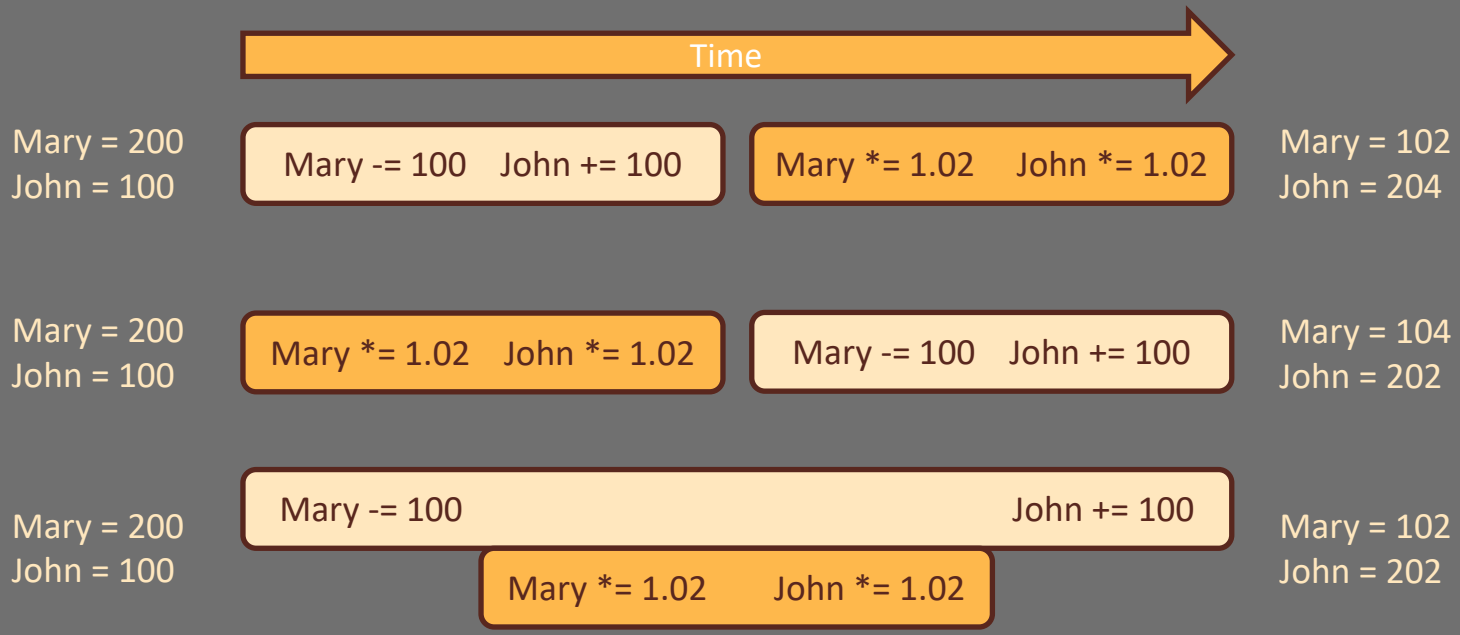
Isolation: Acceptable because result corresponds to a serial order – serializable.

Why interleave?

- Fast transactions shouldn’t have to wait for slow transactions.

- Using parallel hardware. Coordinating transactions adds overhead.

Non-serializable schedules



Isolation: Unacceptable because result doesn't correspond to a serial order – not serializable.



Interleaving & ACID

DBMS can interleave transactions.

… as long as consistency & isolation maintained.